

Home-work 3

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Abstract

This report details the end-to-end data engineering pipeline designed to process, sanitize, and model financial transaction records for the fiscal year 2024. It outlines the strategic procedures undertaken for rigorous data quality remediation—specifically addressing formatting anomalies, handling missing master data, and standardizing geographical taxonomies. Furthermore, the document explores the architectural evolution from a traditional Star Schema to a highly optimized Denormalized Flat Table, engineered specifically to power dynamic and performant frontend analytics via a Streamlit dashboard.

1 Introduction

This report documents the comprehensive data engineering workflow developed to process and analyze financial transaction datasets spanning the fiscal year 2024. The primary objective of this initiative was to transform fragmented, inconsistent raw CSV extracts into a cohesive, high-fidelity analytical dataset.

The project was executed in two primary phases. The first phase focused on rigorous data quality remediation, employing programmatic techniques using Python and Pandas to resolve structural inconsistencies, eliminate phantom records, and enforce strict master data integrity. The second phase involved the strategic design of the database architecture. By deliberately pivoting from a normalized schema to a denormalized flat structure, the data model was heavily optimized for read-intensive operations, ensuring seamless and rapid interactions within the final Streamlit executive dashboard. Ultimately, this pipeline guarantees that the resulting business intelligence is both accurate and highly responsive.

2 Data Quality Remediation

Financial transaction data is highly susceptible to formatting errors, especially when aggregated from multiple heterogeneous sources or exported from spreadsheet applications. During the initial extraction and loading phases, a thorough data profiling process revealed several critical data quality issues.

These anomalies had to be systematically identified and resolved to prevent ETL (Extract, Transform, Load) pipeline failures and to ensure the absolute accuracy of downstream reporting. The following subsections detail the specific challenges encountered and the programmatic solutions implemented to mitigate them.

2.1 Formatting Anomalies & Phantom Rows

The raw CSV files exhibited formatting anomalies, including invisible trailing whitespaces in column headers and completely empty 'phantom' columns (e.g., Unnamed: 5). Programmatically, these were resolved using string manipulation functions (e.g., `str.strip()`) and column dropping techniques in Pandas.

Furthermore, empty rows at the end of the transaction files—often artifacts of spreadsheet software exports—were identified and removed dynamically. This was achieved by enforcing a non-null constraint on the critical Symbol column using the `dropna(subset=['Symbol'])` method, guaranteeing that only valid transactions were processed.

2.2 Inconsistencies & Master Data

Geographic Naming Inconsistencies: Country names varied significantly across datasets (e.g., "United Kingdom of Great Britain and Northern Ireland" vs. "United Kingdom", alongside character encoding anomalies with "Türkie" or "Taiwan"). A standardized dictionary mapping was applied universally to both the geography dimension and the symbol dimension datasets. This harmonization prevented the generation of NaN values during relational table merges.

Missing Master Data: An integrity check revealed that approximately 18 ticker symbols were present in the transaction history but completely missing from the master symbol directory. To preserve financial transaction history without losing data via inner joins, "stub records" were dynamically generated. These missing tickers were appended to the dimension table and explicitly assigned to an "Unknown" sector, industry, and country.

Case Sensitivity: Inconsistencies in column casing (e.g., Date vs. date, Transaction-Type vs. transactiontype) caused pipeline execution failures. This vulnerability was mitigated by coercing a strict lowercase naming convention across all sanitized datasets using Pandas' `str.lower()` attribute on the dataframe columns.

3 Database Architecture

The initial architectural approach involved a traditional Star Schema, characterized by the creation of distinct dimension tables (e.g., `dim_time`, `dim_symbol`, `dim_geo`) linked via surrogate keys to a central fact table. However, to optimize for read-heavy operations and dynamic frontend filtering (e.g., via Streamlit), the architecture was deliberately pivoted to a Denormalized Flat Table.

The final consolidated fact table was enriched directly with descriptive attributes, including year, quarter, transactiontype, sector, and country. This denormalization strategy significantly improved query readability and performance by eliminating the need for complex, repetitive `.merge()` operations in Pandas, while also preventing inadvertent column duplication during multi-table joins.

3.1 Modeling Choices

The initial architectural approach involved a traditional Star Schema, characterized by the creation of distinct dimension tables (e.g., `dim_time`, `dim_symbol`, `dim_geo`) linked

via surrogate keys (`date_key`, `symbol_key`) to a central fact table. Time-based dimensions were programmatically extracted using Pandas datetime properties (e.g., `dt.year`, `dt.quarter`).

However, to optimize for read-heavy operations and dynamic frontend filtering (e.g., via Streamlit), the architecture was deliberately pivoted to a Denormalized Flat Table. Through a series of `.merge()` operations on the surrogate keys, the final consolidated fact table was enriched directly with descriptive attributes, including year, quarter, transactiontype, sector, and country.

To maintain efficiency in memory footprint, redundant columns generated during the merge process (such as duplicate symbol names or transaction type keys) were dropped using list comprehensions and targeted `.drop()` functions. This denormalization strategy significantly improved query readability and frontend performance by eliminating the need for complex, on-the-fly table joins during dashboard execution.

3.2 Schema Definition and Table Hierarchies

Prior to denormalization, the logical data model was designed as a **Star Schema**. This foundational structure separates the quantitative events from their descriptive context, establishing clear hierarchies and relationships across the dataset.

Dimension Tables (Context & Hierarchies): The descriptive data was segmented into four primary dimension tables, each assigned a unique integer surrogate key for efficient joining:

- **dim_time:** Captures the temporal hierarchy of transactions. Granularity rolls up from exact *Date* → *Day* → *Month* → *Quarter* → *Year*. It uses an integer representation of the date (YYYYMMDD) as its `date_key`.
- **dim_geo:** Defines the geographical taxonomy, rolling up from *Country Name* → *Sub-region* → *Region*, linked via the `geo_key`.
- **dim_symbol:** Contains the master data for traded assets, detailing the *Symbol* (ticker), *Sector*, and *Industry*, linked via the `symbol_key`.
- **dim_type:** Categorizes the nature of the transaction (e.g., BUY, SELL), linked via the `type_key`.

Fact Table (`fact_transactions`): Positioned at the center of the schema, the fact table records the core transactional events. It stores all measurable metric (units traded) and maintains one-to-many relationships with the dimension tables via the aforementioned surrogate foreign keys.

3.3 Analytical Output Testing

To validate the structural integrity and efficiency of the denormalized model, a series of complex aggregations were executed. The pipeline successfully processed the analytical queries, demonstrating the model's capacity to handle multi-dimensional grouping.

Here are reported the outputs of queries:

3.3.1 What are the top 5 sectors by number of SELL transactions in US during 2024?

The Technology sector is the undisputed leader in Monday sales volume. At approximately 75,000 units sold, it drastically outperforms the other sectors.

Healthcare holds the second position with roughly 26,000 units sold, followed by Communication Services in third place with nearly 20,000 units.

The most striking feature of the graph is the massive gap between first and second place. The Technology sector's Monday sales volume is nearly three times higher than that of the Healthcare sector, indicating highly concentrated trading or sales activity in tech at the start of the week.

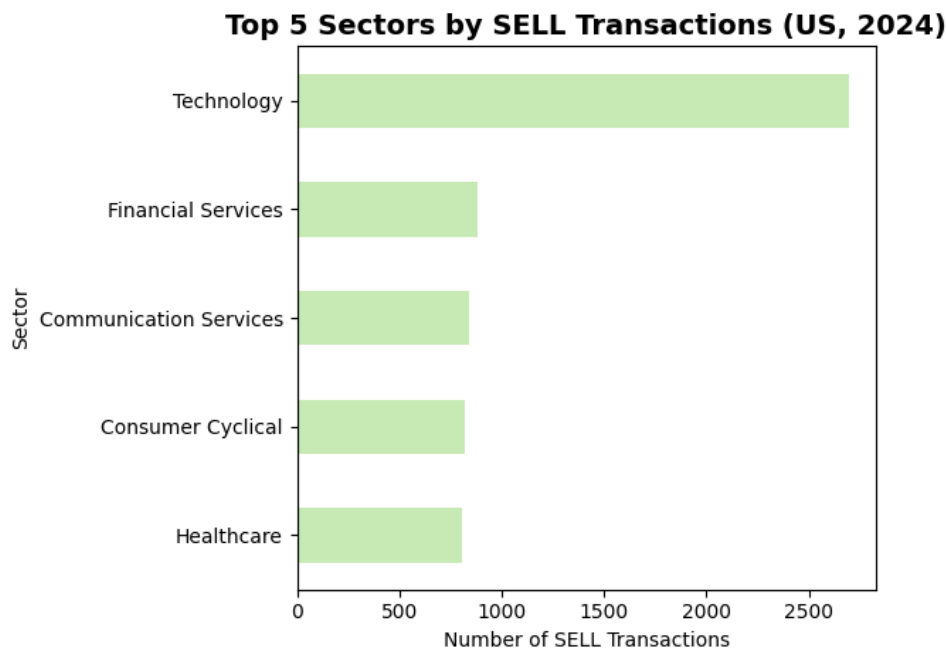


Figure 1:

3.3.2 What are the top 5 industries by number of BUY transactions in Q4 of 2024?

Filtering directly the `dim_type` and `tim_geo` and using them to get exactly the information about the subset of interest; and at the end grouping by the industries. Semiconductors appears in the first place in terms of numbers of BUY transaction in the last three months of 2024, in the second place there is the Internet Content & Information industry, meanwhile Internet Retail, Software - Infrastructure and Diagnostics & research shows the same number of transactions around 65 and 90. Due to the hype and high demand of software services we couldn't help but expect Semiconductors to be the giant compared to the other sector. In fact the other industries to stay alive need the hardware structure behind.

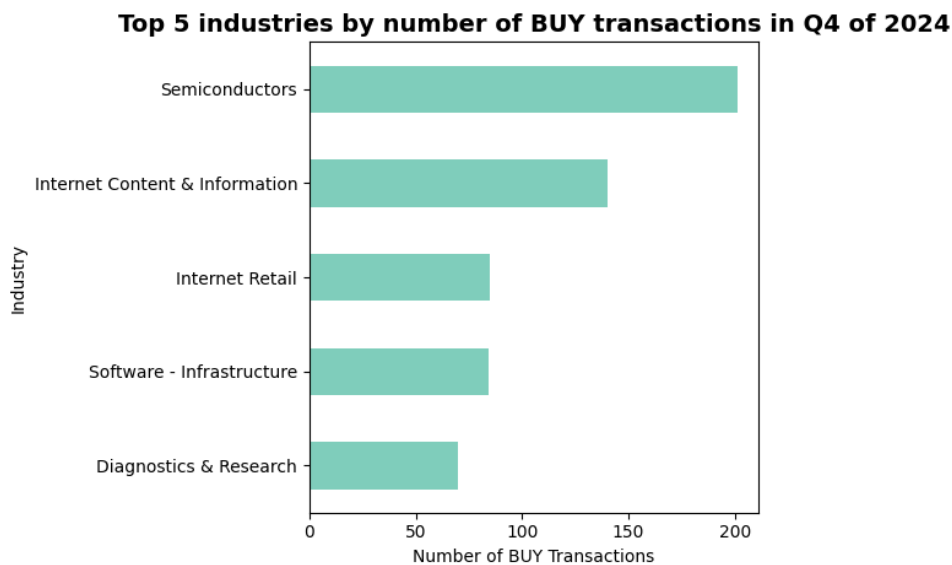


Figure 2:

3.3.3 Rank all quarters of 2024 buy total number of transactions

Looking at the histogram the first three months of the 2024 are the most active in absolute in terms of ferment and activity. Also this time i used the filter dimentions : time(2024) and type_of_transaction(BUY+SELL), and after apply the merge with fact_table only on the filtered subset and obtain the other information.

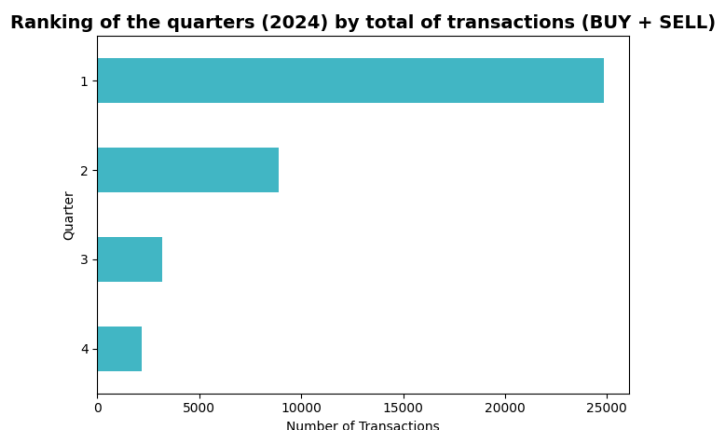


Figure 3:

3.3.4 What are the top 5 sectors by total units traded (BUY + SELL) in Q3 of 2024?

The Technology sector dominates the chart, being the only sector to surpass the 50,000-unit mark. [first place]

Consumer Cyclical follows in second place, with trading volume approaching 40,000 units.

Financial Services and Communication Services hold the third and fourth positions, respectively. Both sectors show similar trading volumes, hovering just slightly above 30,000 units.

Lowest of the Top 5: Healthcare ranks fifth, with its trading volume landing just below the 30,000-unit threshold.

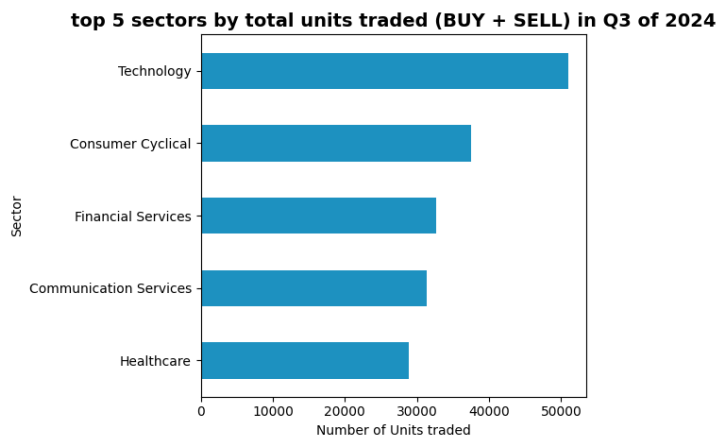


Figure 4:

3.3.5 What are the top 10 symbols by number of transactions (BUY + SELL) in 2024?

ARM clearly dominates the chart. It is the only symbol to cross the 2,000-transaction threshold, sitting at roughly 2,250 transactions for the year.

The upper half of the chart shows massive trading interest in semiconductor and hardware manufacturing. Behind ARM, AMD and TSM secure strong positions in the top four.

There is a significant gap between the top four actively traded symbols and the rest of the list. After TSM, the transaction volume drops steeply, cutting nearly in half.

The remaining symbols—EH, ARDX, GOOG, APP, MSFT, and AMZN—are grouped closely together. They all fall within a much narrower range, logging between approximately 700 and 900 transactions each. Interestingly, widely recognized mega-cap tech giants like Microsoft (MSFT) and Amazon (AMZN) sit at the very bottom of this top 10 list.

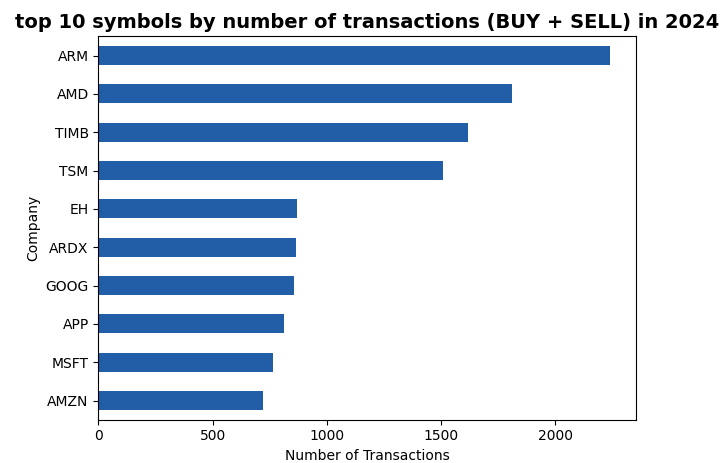


Figure 5:

3.3.6 What are the top 3 sectors by total number of units sold on Mondays across 2024?

The Technology sector is the undisputed leader in Monday sales volume. At approximately 75,000 units sold, it drastically outperforms the other sectors.

Healthcare holds the second position with roughly 26,000 units sold, followed by Communication Services in third place with nearly 20,000 units.

The most striking feature of the graph is the massive gap between first and second place. The Technology sector's Monday sales volume is nearly three times higher than that of the Healthcare sector, indicating highly concentrated trading or sales activity in tech at the start of the week.

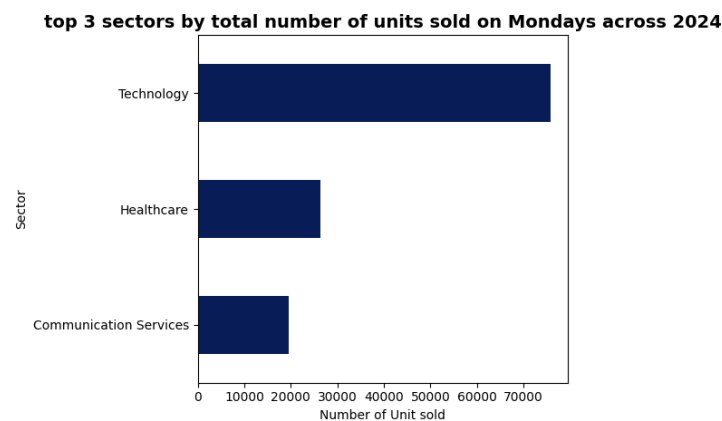


Figure 6: